# DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT

## Calculus and Optimization

Maxima and Minima

Lecture No. 01



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# Recap of previous lecture



Topic

MEAN VALUE THEOREMS

# Topics to be Covered



**Topic**

1. Lagrange's M.V.Th for Integrals
2. Maxima - Minima (Part 1)
   → Increasing & Decreasing func$^{n}$

## (1) Lagrange's M.V.Th

Let $f(x)$ is defined in $[a,b]$ s.t (i) $f(x)$ is Cont in $[a,b]$, (ii) $f(x)$ is Diff in $(a,b)$ then ∃ at least one $c$ in $(b/n)$ $a$ & $b$ for which




$A(a, f(a))$, $B(b, f(b))$

Tangent at $c$ is $\parallel^{el}$ to chord $AB$

i.e. Slope of tangent = Slope of $AB$

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

Note:- Converse of LMVTh is not necessarily True.

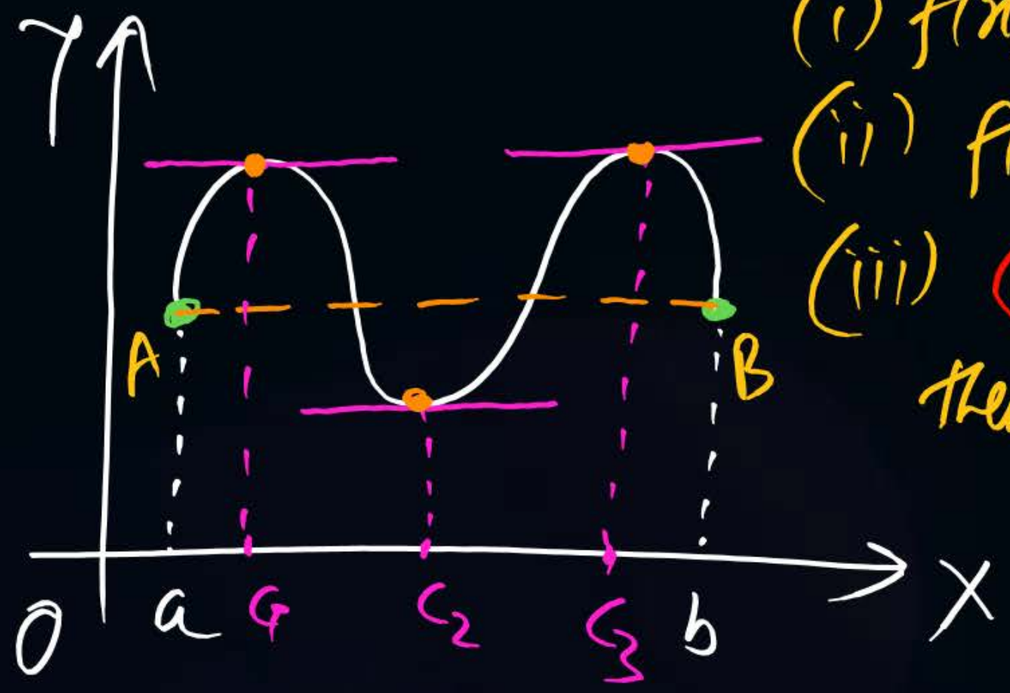
## (2) Rolle's M.V.Th →

Let $f(x)$ is defined in $[a,b]$ s.t

(i) $f(x)$ is Continuous in $[a,b]$

(ii) $f(x)$ is Diff in $(a,b)$

(iii) $f(a)=f(b)$




then $\exists$ at least one point $c$ in $(b/w)$ $a$ & $b$ for which

Tangent is Horizontal

or tangent is $\perp^r$ to X axis

or $f'(c)=0$

(3) **Cauchy's M.V.Th** :- Let $f(x)$ & $g(x)$ are two func$^n$ defined in $[a,b]$ s.t

(i) Both $f(x)$ & $g(x)$ are continuous in $[a,b]$

(ii) " " " " differentiable in $(a,b)$

(iii) $g'(x) \neq 0 \ \forall \ x \in (a,b)$

then $\exists$ at least one $c$ in $(b/n)$ $a$ & $b$ for which

$$\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$$

(Ratio of Difference of $f(x)$ & $g(x)$ at $a$ & $b$) = (Ratio of slope of tangent of $f(x)$ & $g(x)$ at $c$)

Qe Consider $f(x)=(x-1)^2+2$ then Verify LMVT in $[0,1]$ (Ans $c=\frac{1}{2}$)

OR

Qe Consider $f(x)=(x-1)^2+2$ then find Coordinates of that point on the curve $f(x)$ whose tangent is $\parallel^{el}$ to the chord joining $(0,3)$ & $(1,2)$

OR

Ans $\left(\frac{1}{2}, \frac{9}{4}\right)$

Qe if $f(x)=(x-1)^2+2$ then find equ$^n$ of tangent which is $\parallel^{el}$ to the chord joining $(0,3)$ & $(1,2)$.

Ans: $4x+4y=11$

OR

Qe for the func$^n$ $y=(x-1)^2+2$, At least, at one point in the interval $[0,1]$, the value of $\frac{dy}{dx}=?$ (Ans $=-1$)

Ex: $f(x)=(x-1)^2+2$ in $[0,1]$ $\begin{cases} a=0, & f(a)=f(0)=3 \\ b=1, & f(b)=f(1)=2 \end{cases}$

(1) i.e. $A(a,f(a))=(0,3)$ & $B(b,f(b))=(1,2)$

→ It is a polynomial func$^n$ of degree 2 (i.e. E-func$^n$) Hence Cont. as well as Diff in $(0,1)$

So By L.M.V.Th, $f'(c)=\dfrac{f(b)-f(a)}{b-a}$

$f'(c)=\dfrac{2-3}{1-0}=-1$

$2(c-1)=-1$

$c=1/2$

$\because$ c lies within in $(0,1)$

Hence LMVT is verified

(2) $c=\frac{1}{2}$ & $f(c)=(c-1)^2+2$

$\Rightarrow f(\frac{1}{2})=(\frac{1}{2}-1)^2+2=\frac{9}{4}$

So Required point $=(c, f(c))=(\frac{1}{2}, \frac{9}{4})$

(3) Slope of tangent at C is $m=f'(c)=-1$

So equ$^n$ of tangent Passing through $(\frac{1}{2}, \frac{9}{4})$ having slope $m=-1$ is

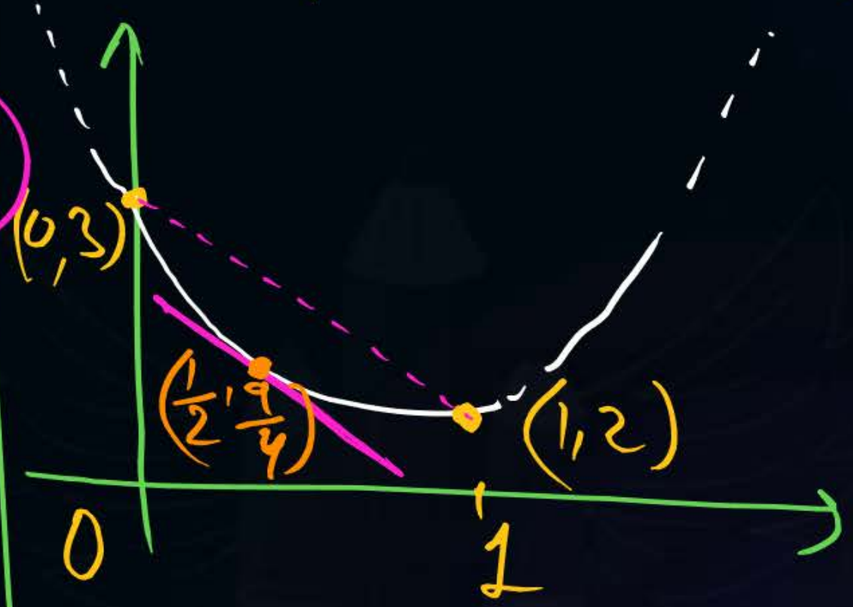
$$y-\frac{9}{4}=(-1)(x-\frac{1}{2})$$

$$4x+4y=11 \quad \underline{\text{Ans}}$$

Analysis: $y=(x-1)^2+2$

or $(y-2)=(x-1)^2$

$Y=X^2$ so vertex $(1,2)$



(4) $y=(x-1)^2+2$, then at least at one point in the interval $(0,1)$ the Value of $\frac{dy}{dx}=$ ?

Sol: By L.M.V.T we know that at $x=c$, $f'(c)=-1$

or $\left(\frac{dy}{dx}\right)_{x=c}=-1$

$\because$ c lies in b/n 0 & 1 Hence we can take our ans as $-1$ Ans

Q: for the func$^n$ $y=5x^2+10x$, At least at one point in $(1,2)$, $\frac{dy}{dx}$ is exactly ?

By L.M.V.T, $f'(c)=\frac{f(2)-f(1)}{2-1}$

ie $f'(c)=\frac{40-15}{2-1}=25$

$\because c\in(1,2)$ so $\left(\frac{dy}{dx}\right)_{x=c}=25$ Ans

(M-II) $y = 5x^2 + 10x$

$$\frac{dy}{dx} = 10x + 10$$

$$\left(\frac{dy}{dx}\right)_{x=c} = 10c + 10 = ????$$ Not able to get unique answer.

& when we apply L-M.V.Th, we will get c as follows.

$$f'(c) = \frac{f(2) - f(1)}{2 - 1} = 25$$ Ans of our Question.

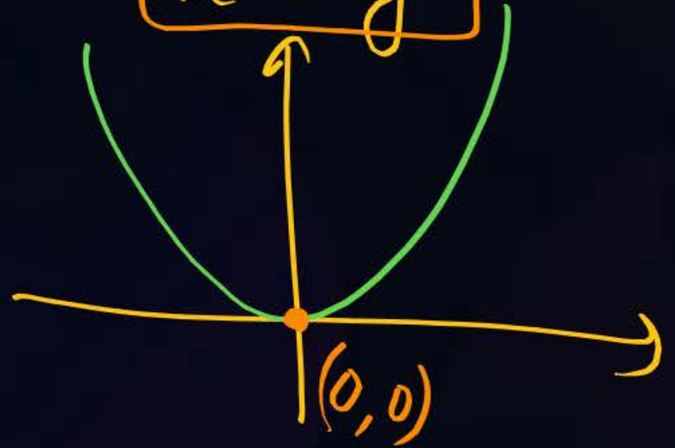
$10c + 10 = 25 \Rightarrow c = \frac{3}{2}$ wastage of time.

Note: ① $y=(x-1)^2+2$, $\frac{dy}{dx}=2(x-1)$

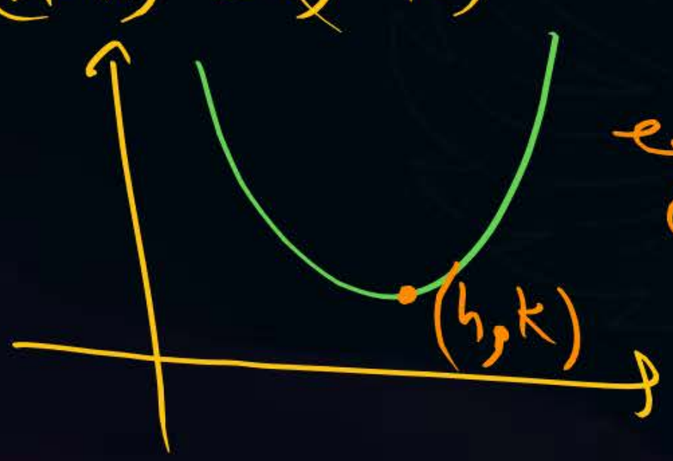
when $x\in(0,1)$ then Various ans are $\frac{dy}{dx}=?$

$\left(\frac{dy}{dx}\right)_{x=\frac{1}{4}}=-\frac{3}{2}$ ✓, $\left(\frac{dy}{dx}\right)_{x=\frac{1}{2}}=-\frac{7}{4}$ ✓, $\left(\frac{dy}{dx}\right)_{x=\frac{3}{4}}=-\frac{1}{2}$ ✓, $\left(\frac{dy}{dx}\right)_{x=\frac{1}{2}}=-1$ ✓

② $x^2=y$




$(x-h)^2=(y-k)$



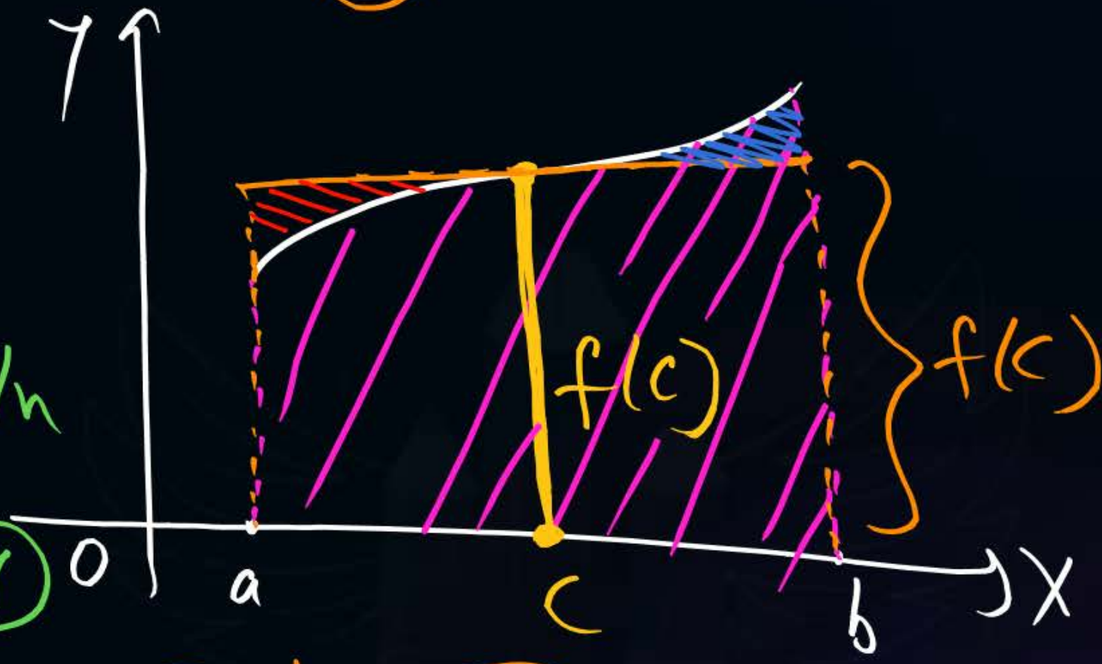

eg Vertex of $(y-2)=(x-1)^2$ is $(1,2)$

# Lagrange's Mean Value Theorem for Integrals →

Average Height

if $f(x)$ is continuous func$^n$ in $[a,b]$ then $\exists\ c$ in b/n $a$ & $b$ s.t

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx$$



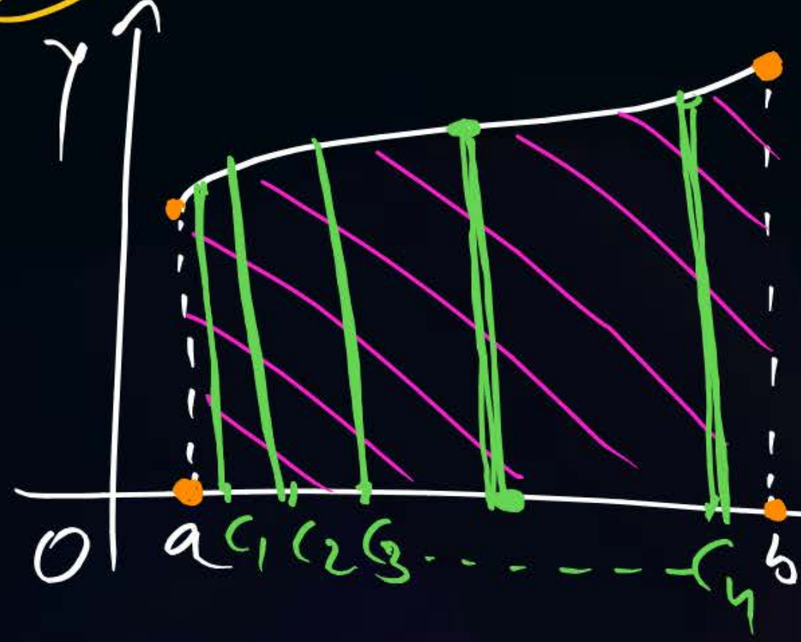

Proof: w.k.t $\int_a^b f(x)\,dx$ = Area under $f(x)$ b/n $a$ & $b$ and X axis — (1)

Area of this Rectangle = length × height

$= (b-a) \times f(c)$ — (2)

By (1) & (2), $(b-a)\times f(c)=\int_a^b f(x)\,dx \Rightarrow \boxed{f(c)=\frac{1}{b-a}\int_a^b f(x)\,dx}$

M-II




Av. Height of $f(x)$ is given as,

$$f(c)=\frac{f(c_1)+f(c_2)+f(c_3)+\cdots+f(c_n)}{n}$$

$$f(c)=\frac{\int_a^b f(x)\,dx}{\text{length of interval}}=\frac{\int_a^b f(x)\,dx}{b-a}$$

i.e Av. Height of Curve $=\frac{1}{b-a}\int_a^b f(x)\,dx$

**Note:** i.e Average Height of Curve b/n a & b is $= f(c)$
& it occurs at $x = c$ where $c \in (a, b)$

**Verification:** Find Av. Height of $y = 2x$ b/n 1 & 6

**M-I**

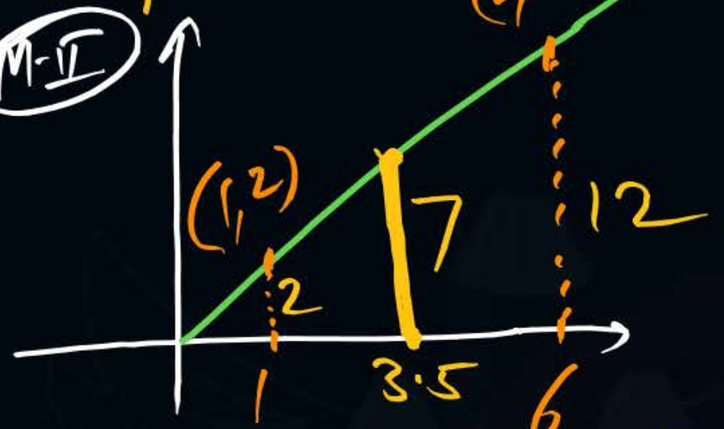
$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$= \frac{1}{6-1}\int_1^6 (2x)\,dx$$

$$\text{Av. Height} = \frac{1}{5}(x^2)_1^6 = \frac{1}{5}(36-1) = 7$$

& it occurs at $2c = 7 \Rightarrow c = 3.5$

**M-II**




i.e when $x \in (1,6)$, $y \in (2,12)$
Hence Average $y = \frac{2+12}{2} = 7 = f(c)$
& it occurs at $c = 3.5$

Q: Find the Average Value of $f(x)=x^2$ b/n 1 & 4 ?

Sol:
$$f(c)=\frac{1}{b-a}\int_a^b f(x)\,dx=\frac{1}{4-1}\int_1^4 x^2\,dx=\frac{1}{3}\left(\frac{x^3}{3}\right)_1^4=\frac{64-1}{9}=7$$

(ii) Also find the Coordinates of that point?

(a) $\pm\sqrt{7}$

(b) $(\sqrt{7},7)$

(c) $(-\sqrt{7},7)$

(d) $(2.5,7)$

$\because f(c)=7$

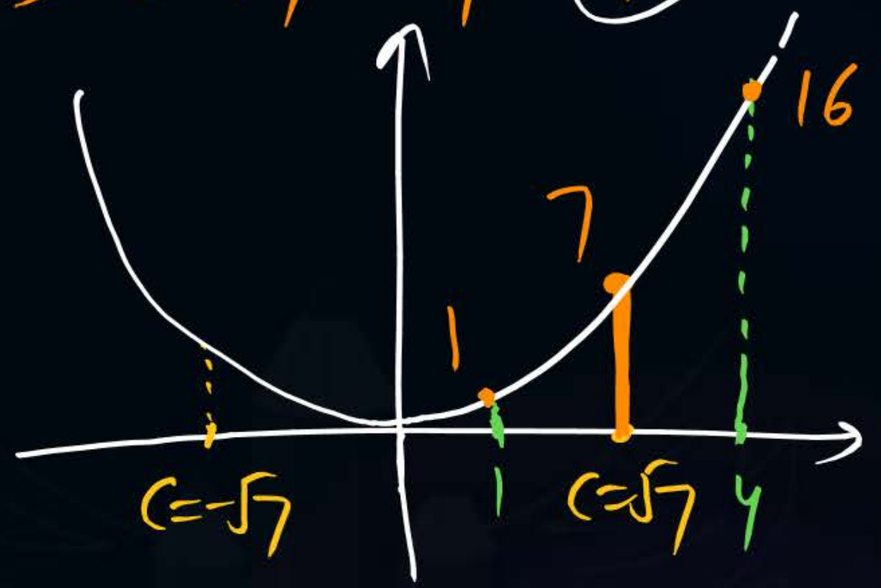
$c^2=7$

$c=\pm\sqrt{7}$

$\because c=-\sqrt{7}\notin(1,4)$

So Neglect it & we Will take only $c=\sqrt{7}$

So Req. Point $=(\sqrt{7},7)$



Q: 2022 The Mean Value (Average Value) of $f(x) = 4x^2 + 3x + 1$ b/n $-1$ & $2$ is ?

Sol:

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx = \frac{1}{2-(-1)}\int_{-1}^{2}(4x^2+3x+1)\,dx$$

$$= \frac{1}{3}\left[4\frac{x^3}{3} + 3\frac{x^2}{2} + x\right]_{-1}^{2}$$

$$= \frac{1}{3}\left[\left\{\frac{4}{3}(8) + \frac{3}{2}(4) + 2\right\} - \left\{\frac{4}{3}(-1) + \frac{3}{2}(1) + (-1)\right\}\right]$$

$$= \text{Ans}$$

# Maxima & Minima (Part-1) → Unit

(Increasing & Decreasing func$^n$) → Chapter

W.K.T that for $y = f(x)$, $\frac{dy}{dx} = f'(x) = \tan\theta$ = Slope of tangent at any Random Point $x$ on $f(x)$

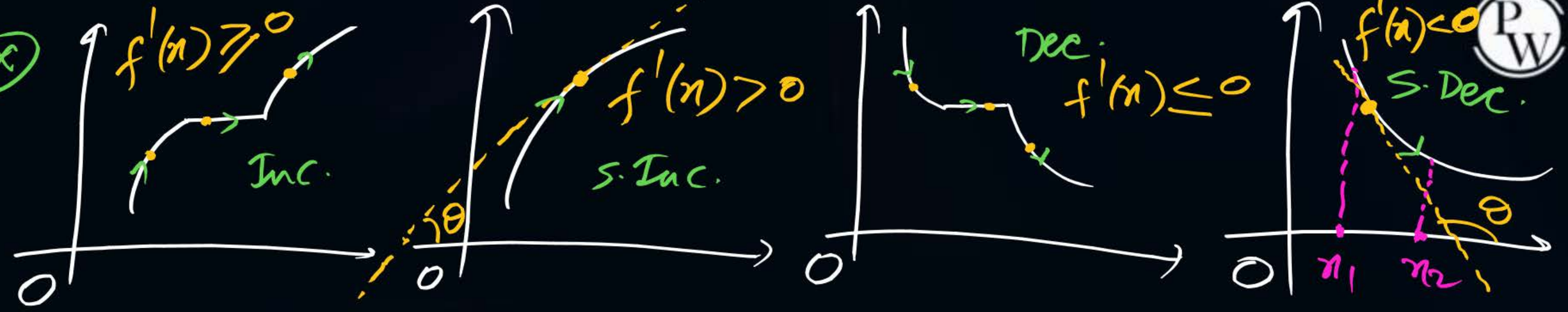



(1) for Acute angle tangent, $f'(x) > 0$

(2) " obtuse " " , $f'(x) < 0$

(3) for Horizontal tangent, $f'(x) = 0$

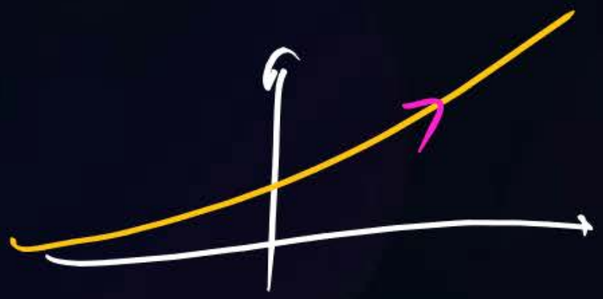
(4) for Vertical Tangent, $f'(x) = D.N.E$

(*) 


eg $f(x_1) > f(x_2)\ \forall\ x_1 < x_2$ ; $x_1, x_2 \in D_f$ then $f(x)$ is S. Dec

eg $f(x_1) \geq f(x_2)\ \forall\ x_1 < x_2$ ; $x_1, x_2 \in D_f$ then $f(x)$ is Decreasing

eg $f(x) = e^x$ is S. Inc



(M-II) $f(x) = e^x$ , $D_f = (-\infty, \infty)$

$f'(x) = e^x$

$\because f'(x) > 0$ always so $f(x)$ is S. Inc.

eg $f(x)=\log_a x$ is S.Inc.

where $a>1$

$D_f=(0,\infty)$

(M-I)



(M-II) $f(x)=\log_a x=\dfrac{\log_e x}{\log_e a}$

$f'(x)=\dfrac{1}{x\log_e a}=\dfrac{1}{(+)(+)}=+ve$

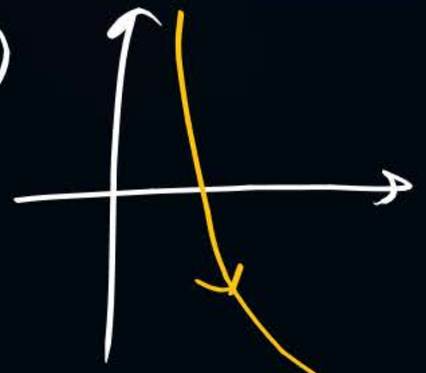
$\because f'(x)>0\ \forall\ x\in D_f \Rightarrow$ $f(x)$ is S.Inc.

eg $f(x)=\log_a x$ is S.Dec.

where $0<a<1$

$D_f=(0,\infty)$

(M-I)



(M-II) $f(x)=\log_a x=\dfrac{\log_e x}{\log_e a}$

$f'(x)=\dfrac{1}{x\log_e a}=\dfrac{1}{(+)(-)}=-ve$

$\because f'(x)<0 \Rightarrow$ $f(x)$ is S.Dec

Note: ① $a > 1$ then $\log_e a = ? = +ve$ Learn

∵ Let $a = 2$, $\log_e 2 = 0.693 = +ve$.

② $0 < a < 1$, $\log_e a = ? = -ve$ Learn.

∵ Let $a = \frac{1}{2}$, $\log_e\left(\frac{1}{2}\right) = \log_e(2)^{-1} = -1 \log_e 2 = -0.6932$

$= -ve$.

Q. $f(x)=\dfrac{e^x}{1+e^x}$ is ?

(a) ✓ Monotonically Increases

(b) ✗ Inc to Max Point & then Dec (i.e Maxima occurs)

(c) ✗ Decreases

(d) ✗ Can't say anything

$f(x)=\dfrac{e^x}{1+e^x}$, $D_f=(-\infty,\infty)$

$$f'(x)=\frac{(1+e^x)(e^x)-e^x(0+e^x)}{(1+e^x)^2}$$

$$f'(x)=\frac{e^x}{(1+e^x)^2}$$

By Common Sense, $f'(x)>0$ always.

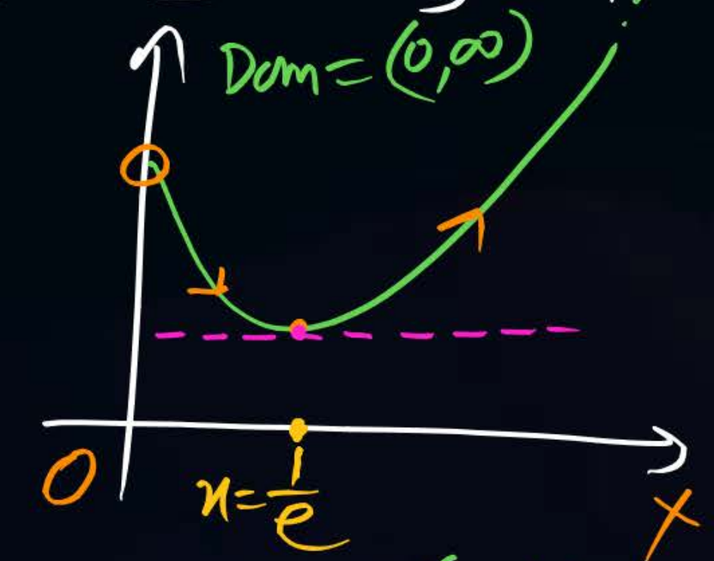
$\Rightarrow$ $f(x)$ is S. Inc. func$^n$.

Note: (1) those func$^n$ which are either S. Inc or S. Dec are called Monotonic func$^n$

(2) while Solving Questions Based on Inc & Dec func$^n$, it is advisable to first write domain of the func$^n$.

**Q:** find the Interval in which $f(x) = x^x$ Increases & Decreases.

**Sol:** (M-I) (using Graph) →




$f(x)$ s. Dec in $\left(0, \frac{1}{e}\right)$

$f(x)$ s. Inc in $\left(\frac{1}{e}, \infty\right)$

(M-II) (W/o using Graph) →

$f'(x) = x^x(1+\log_e x)$

Put $f'(x) = 0$

$x^x(1+\log_e x) = 0$

$\because x^x \neq 0,\ 1+\log_e x = 0$

$\log_e x = -1$

$x = e^{-1} = \frac{1}{e}$

is Turning point of $f(x)$ is $x = \frac{1}{e}$

Now we will check sign of

$f'(x) = x^x(1+\log_e x)$

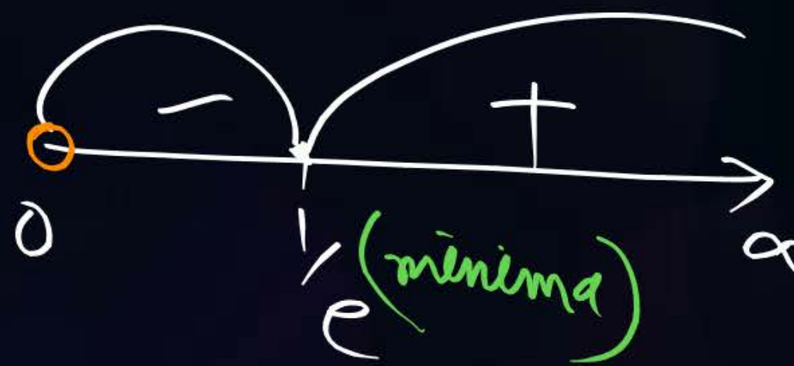
to find the Required interval in which $f(x)$ Increases & Decreases.

P.T.O

$$f'(x) = x^x(1+\log_e x)$$

$f'(e) = e^e(1+\log_e e) = +ve(1+1) = +ve$

$f'\left(\frac{1}{e^2}\right) = \left(\frac{1}{e^2}\right)^{\frac{1}{e^2}}\left[1+\log_e\left(\frac{1}{e^2}\right)\right] = +ve\left[1+\log_e e^{-2}\right] = +ve\left[1-2(1)\right] = -ve$




Sign of $f'(x)$

i.e. $f'(x) < 0 \ \forall\, x \in \left(0, \frac{1}{e}\right)$

so $f(x)$ is S.Dec in $\left(0, \frac{1}{e}\right)$

& $f'(x) > 0 \ \forall\, x \in \left(\frac{1}{e}, \infty\right)$

i.e. $f(x)$ is S.Inc in $\left(\frac{1}{e}, \infty\right)$

Q. Find the Interval in which $f(x)=(x^2-4)^2$ Increases & Decreases?

Sol: $f(x)=(x^2-4)^2$ ; $(-\infty,\infty)$

$f'(x)=2(x^2-4)(2x)$

$f'(x)=4x(x-2)(x+2)$

T. Points of $f(x)$ are

$f'(x)=0$

$4x(x+2)(x-2)=0$

$x=-2,0,2$




$f'(-3)=(-ve)(-ve)(-ve)=-ve$

$f'(-1)=(-ve)(-ve)(+ve)=+ve$

$f'(1)=(+ve)(-ve)(+ve)=-ve$

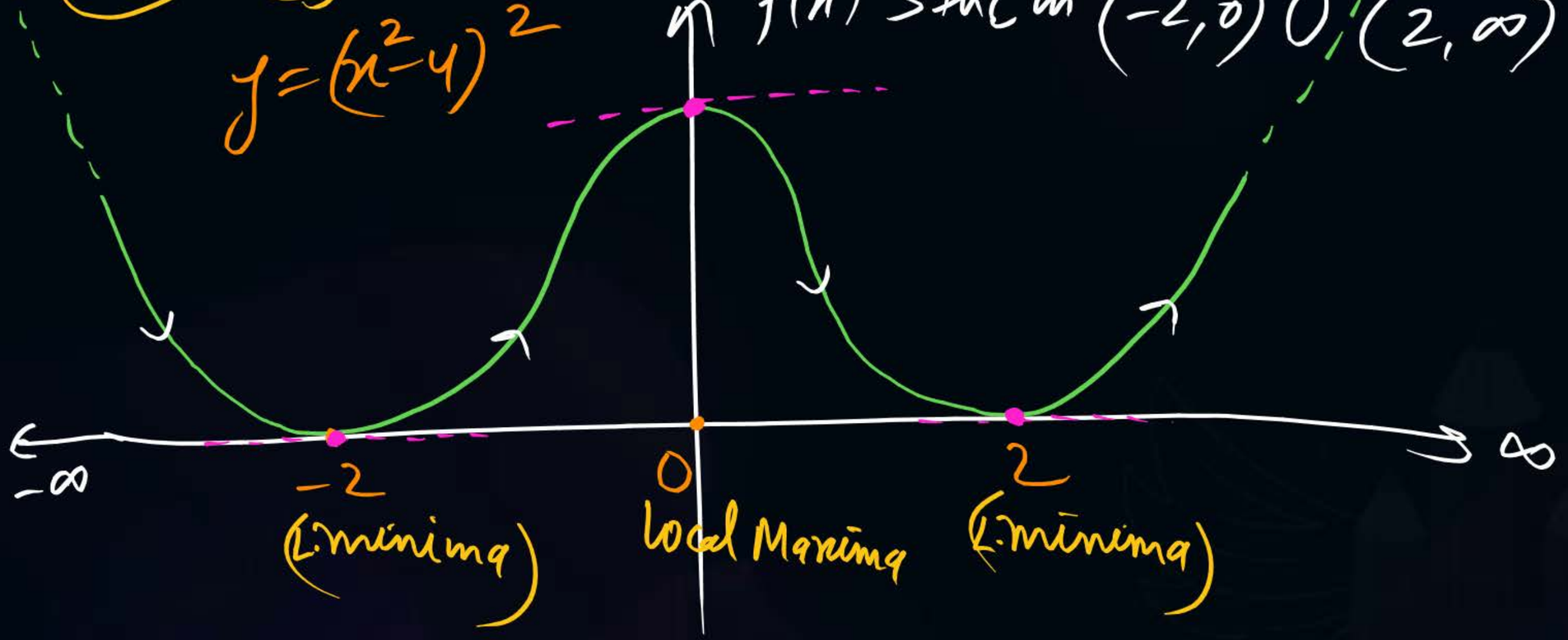
$f'(3)=(+ve)(+ve)(+ve)=+ve$

$f(x)$ S.Inc in $(-2,0)\cup(2,\infty)$

$f(x)$ SD in $(-\infty,-2)\cup(0,2)$

**M-II (using Graph) →**

$y = (x^2 - 4)^2$




$f(x)$ S.Dec in $(-\infty, -2) \cup (0, 2)$

$f(x)$ S.Inc in $(-2, 0) \cup (2, \infty)$

(HW) Find the Intervals of Increasing & Decreasing for $y = \frac{\log_e x}{x}$

Domain $= (0, \infty)$

Sol:

# THANK - YOU

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